



Trace the Bug

Quick facilitation notes & answer key

Goal

Students TRACE a loop one repeat at a time to find where it goes wrong (C3.2), then name the fix — covering wrong count, too many, overshoot, and wrong turn.

Curriculum (Ontario Math 2020, Strand C3)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential, concurrent, and repeating events.

C3.2 — read and alter existing code, including code that involves sequential, concurrent, and repeating events, and describe how changes to the code affect the outcomes.

Materials

Python Turtle / a block app projected; the printed worksheet + a pencil. A trace table on the board (repeat # → what happened) helps.

Run it (about 30 min)

1. Model a trace table: repeat 1 → side 1, repeat 2 → side 2, ... for the range(3) square.
2. The trace ends at 3 sides — a square needs 4. Name bug + fix together.
3. Students do Exercises 1–4 (Ex 4 is the stretch), solo or in pairs.
4. Share answers; land the big idea: trace to the end, then compare to the goal.

Answer key (checked by the run-gate)

Ex 1 — 3 sides; the bug is range(3); fix to range(4).

Ex 2 — 8 jumps; fix to repeat 2 ($2 \times 2 = 4$).

Ex 3 — 5 sides (one too many / overshoot); fix to range(4).

Ex 4 (challenge) — the TURN is wrong (right 60); a square needs right 90.

Watch for & differentiate

Common slip: guessing instead of tracing each repeat.

Support: fill the trace table together for the first repeat, then students continue.

Extend: trace range(6) f50 r60 — a hexagon — and confirm it closes.

Success looks like

The student traces a loop to the end, compares the result to the goal, and names both the bug and the fix (Ex 1–4).